ONLINE SHOPPING SYSTEM

Project Description Report

Group Members

1. Name: AHMED GAMALELDIN Student ID: 202512523
2. Name: MOHAMMED ALFATIH Student ID: 202513073

Institution: Kigali Independent University (ULK)

Project Description Report

1. Introduction

The Online Shopping System is a proposed information system that allows customers to browse products, place orders, and record payments online. It helps organise information about customers, products, shopping carts, orders, and payments.

2. Problem Statement

Managing product details, customer information, and orders manually can be time-consuming and may lead to errors. Customers may also need to visit physical shops to find and purchase products. The proposed system aims to organise these activities in one system.

3. Project Purpose

The purpose of this project is to model an Online Shopping System using a UML Class Diagram. The model identifies the main classes, their attributes, methods, and relationships.

4. Project Scope

Functions included:

• Managing customer information.

• Displaying and maintaining product details and stock.

• Adding and removing products from shopping carts.

• Calculating cart and order totals.

• Placing and tracking orders.

• Recording payment information and payment status.

Functions excluded:

• Real-time integration with banks or external payment gateways.

• Physical product delivery management.

• Building a fully functional website or mobile application.

This assignment focuses on system analysis and UML modelling, not implementation of a complete application.

5. Main Classes and Responsibilities

• Customer: stores customer information and manages customer profile details.

• Product: stores product details, price, and available stock.

• ShoppingCart: stores a customer's selected items and calculates the cart total.

• CartItem: records a product, quantity, and unit price in a shopping cart.

• Order: stores order date, status, total amount, and order operations.

• OrderItem: records a product, quantity, and unit price within an order.

• Payment: records payment amount, method, date, and status.

6. System Requirements

The system should:

• Store and update customer information.

• Store product information and available stock.

• Add and remove cart items.

• Calculate item subtotals and the cart total.

• Create orders and track their status.

• Record payment details and payment status.

7. Conclusion

The proposed Online Shopping System provides a structured model for common online shopping activities. The UML Class Diagram illustrates the classes and how they relate to one another.